1. The table gives the initial monthly premiums for males and females aged 50-54 as a function of the amount of life insurance. Complete parts a to d.

Life Insurance Amount	Male Premium	Female Premium
$20,000	$11.32	$5.13
25,000	14.15	6.42
50,000	28.29	12.83
75,000	42.44	19.25
100,000	56.58	25.67

a. If the function f(x) is the premium for males for life insurance amount x, in dollars, find f(100,000) and f(75,000) in dollars.

f(100,000) = ☐ (Type an integer or a decimal.)

f(75,000) = ☐ (Type an integer or a decimal.)

b. If the function g(x) is the premium for females for life insurance amount x, in dollars, find g(100,000) and g(50,000) in dollars.

g(100,000) = ☐ (Type an integer or a decimal.)

g(50,000) = ☐ (Type an integer or a decimal.)

c. For what value of x is f(x) = 42.44?

x = ☐ (Type an integer or a decimal.)

d. Is f(x) greater than g(x) for all x-values?

○ Yes

○ No

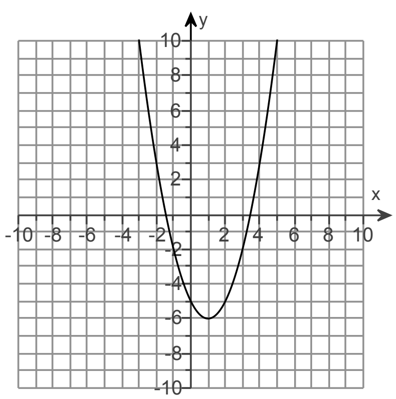
2. Watch the video and then solve the problem given below.

Click here to watch the video.[1]

For each of the functions y = f(x) described below, find f(− 1).

(a) $y = 8 - 4x^2$

(b)

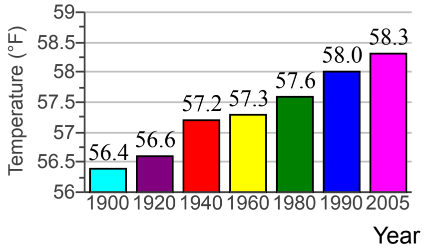


(a) f(− 1) = ☐

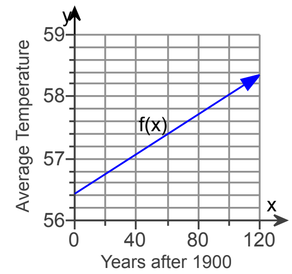
(b) f(− 1) = ☐

1: https://mediaplayer.pearsoncmg.com/assets/CA5HFY010105

*3. The bar graph shows the average temperature for a certain city, in degrees Fahrenheit, for seven selected years.



The data can be modeled by the linear function $f(x) = 0.016x + 56.43$, where f(x) is the average temperature, in degrees Fahrenheit, x years after 1900. The graph of f is shown to the right. Complete parts (a) through (b).



a. Find and interpret f(90). Use the equation for f(x) to identify this information as a point on the graph.

The point is ______.
(Type an ordered pair.)

Interpret this information.

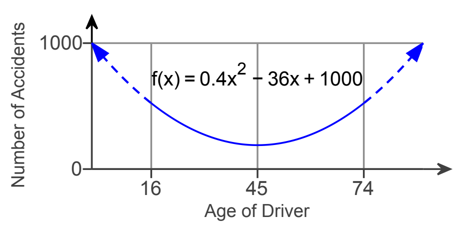
The average temperature in the city in the year ______ was ______ °F.

b. Does f(90) overestimate or underestimate the actual data shown by the bar graph? By how much?

The value f(90) (1) ______ the data in the bar graph by ______ °F.

(1) ◯ overestimates
◯ underestimates

*4. The function models the number of accidents, f(x), per 50 million miles driven as a function of a driver's age, x, in years, where x includes drivers from ages 16 through 74, inclusive. The graph of f is shown. Find and interpret f(66). Identify this information as a point on the graph of f.



Using the equation for f, find f (66).

f (66) = ______ (Simplify your answer. Type an integer or a decimal.)

Interpret f (66).

For ______-year-old drivers, there are ______ accidents per 50 million miles driven.

Identify this information as a point on the graph of f.

______ (Type an ordered pair.)